IMPaCT-Data Biomedical Cloud

A national implementation of the European Genomic Data Infrastructure (GDI)

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European Genomic Data Infrastructure

IMPaCT Data

The European Genomic Data Infrastructure (GDI) project aims to establish a federated and secure platform facilitating access and analysis of genomic, phenotypic, and clinical data across Europe. This initiative operates through a network, where each node represents an European country responsible for implementing the required software stack to enable data sharing and processing within this infrastructure.

The IMPaCT-Data Biomedical Cloud is the Spanish national implementation of the European Genomic Data Infrastructure (GDI). It is being established to provide a scalable and flexible analysis environment, enabling the integration, management and analysis of clinical, genomic and medical imaging data available within the Spanish National Precision Medicine Infrastructure associated with Science and Technology (IMPaCT).

DATA DISCOVERY

Data catalogue

A central User Portal displays the metadata of the datasets shared through FAIR Data Point by the different nodes

FAIR DATA POINT

Beacon network

Genomic and phenotypic queries can be made to find out the datasets of interest using GDI/IMPaCT-Data Beacon Network

DATA ACCESS

Authorization

Access to each dataset should be requested through central REMS, which will issue a visa to be included in the user GA4GH passport

Authentication

Federated authentication system through LS-AAI, which is included as an IdP in the local Secure Processing Environment

KEYCLOAK

LS LOGIN

Authorization

A central user administration system manages authorization assertions provided by nodes as GA4GH passport claims

Authentication

Federated authentication system with a common central server to enable Single Sign-On with different IdPs, including LS-AAI

KEYCLOAK

LS LOGIN

DATA STORAGE

EGA

Datasets are stored in the EGA archive together with their associated metadata

EUROPEAN GENOME-PHENOME ARCHIVE

Security

EGA datasets are encrypted using Crypt4GH and user's private key is used during the computation to process them

DATA PROCESSING

VRE

A VRE is the solution to provide a Secure Processing Environment where the user can perform an analysis on a dataset

Virtual Research Environment

Galaxy

Galaxy is an open-source, community-driven VRE for bioinformatics that can perform distributed data analysis thanks to the Pulsar server application

Galaxy PROJECT

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European Genomic Data Infrastructure

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IMPaCT Data

IMPaCT-Data documentation

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